

Lecture #7 / 24 Sept 2026

- Homework due next Tuesday @ 3:30 PM
- Recap: Laplace Transform
→ why Laplace? Algebra!

- Go back: Partial Fraction Exp.

Hard way: evaluating all coeff's and then solving the system of linear equations

* Cover Up Method

$$Y(s) = \frac{(s+2)(s+4)}{s(s+1)(s+3)} = \frac{C_1}{s} + \frac{C_2}{s+1} + \frac{C_3}{s+3}$$

solve for C_i

$$s Y(s) \Big|_{s=0} = C_1 = \frac{(s+2)(s+4)}{(s+1)(s+3)} \Big|_{s=0} = \frac{8}{3}$$

$$(s+1) Y(s) \Big|_{s=-1} = C_2 = \frac{(s+2)(s+4)}{s(s+3)} \Big|_{s=-1} = -\frac{3}{2}$$

$$(s+3) Y(s) \Big|_{s=-3} = C_3 = \frac{(s+2)(s+4)}{s(s+1)} \Big|_{s=-3} = -\frac{1}{6}$$

cover up method
 $C_i = (s-p_i) F(s) \Big|_{s=p_i}$
 "cover up"
 $P_i = \text{the pole location}$

This only works for signals with distinct poles!

what about $Y(s) = \frac{1}{s^2(s+1)^2}$?

Read FPE Appendix A.

Go back:

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s \cdot F(s)$$

where does this come from?

$$\text{Final Value: } f(\infty) = \underbrace{f(0)}_{\text{initial value}} + \underbrace{\int_0^{\infty} \dot{f}(t) dt}_{\text{sum of all changes to } f \text{ as it goes to } \infty}$$

$$\mathcal{L}\{\dot{f}\} = sF(s) - f(0) \leftarrow$$

$$sF(s) = \mathcal{L}\{\dot{f}\} + f(0)$$

$$\mathcal{L}\{\dot{f}\} = \int_0^{\infty} \dot{f}(t) e^{-st} dt$$

$$sF(s) = f(0) + \int_0^{\infty} \dot{f}(t) e^{-st} dt$$

what happens when $s \rightarrow 0$?
They're equal!

Where are we?

Where are we going?

- 1) Model systems using ODE's
- 2) Express ODE's in the s-domain (using Laplace)
- 3) Express signals as (infinite) series of complex exponentials.
- 4) Find the output of ANY LTI system given ANY input

Ch. 3
in FPE

- 5) Converting s-domain rep's into time-domain spec's
- 6) Block diagram algebra
- 7) Stability

Ch 4
in FPE

- 8) First look at feedback and it does to system's response
→ SSE, PID, FW/FB

Design!

Ch 5

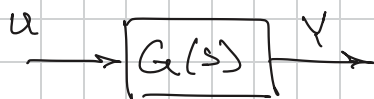
- 9) Root Locus

Ch 6

- 10) Frequency Domain Analysis

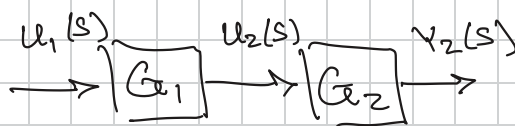
Block diagrams

①



$$\frac{Y(s)}{u(s)} = G(s)$$

②

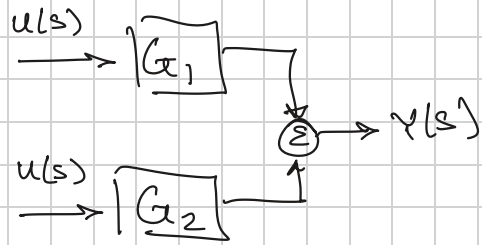
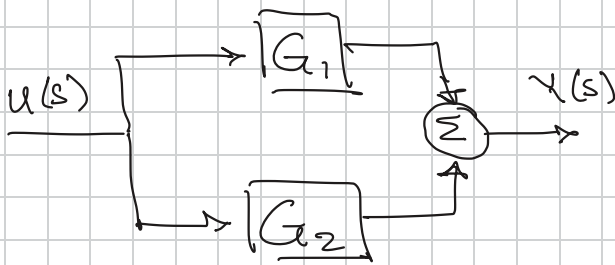


$$\frac{Y_2}{u_1} = G_1 \cdot G_2$$

$$\frac{Y_2}{u_2} = G_2(s)$$

$$\frac{u_2}{u_1} = G_1(s)$$

$$\left(\frac{Y_2}{u_2} \right) \left(\frac{u_2}{u_1} \right) = G_1 \cdot G_2$$



$$\frac{Y(s)}{u(s)} = G_1(s) + G_2(s)$$



$$\frac{Y(s)}{R(s)} =$$

$$\textcircled{1} R(s) - Y(s)G_2(s) = E(s)$$

$$\textcircled{2} Y(s) = E(s) \cdot G_1(s)$$

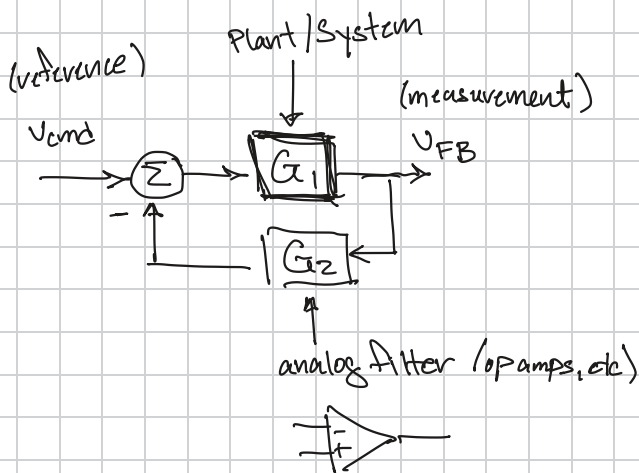
$$\textcircled{3} Y(s) = [R(s) - Y(s)G_2(s)]G_1(s)$$

$$= R(s)G_1(s) - Y(s)G_1(s)G_2(s)$$

$$Y G_1 G_2 + Y = R G_1$$

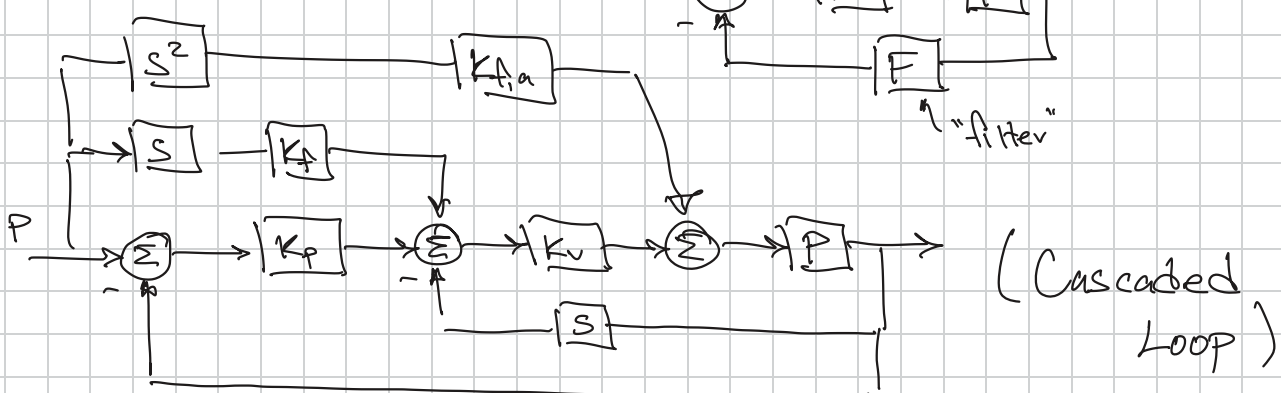
$$Y(1 + G_1 G_2) = R G_1$$

$$\frac{Y}{R} = \frac{G_1}{1 + G_1 G_2}$$



$$\frac{V_{FB}}{V_{cmd}} = \frac{G_1}{1 + G_1 G_2}$$

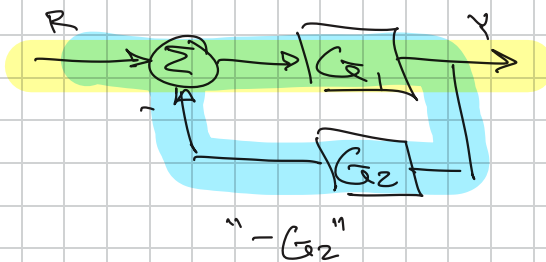
In the future...



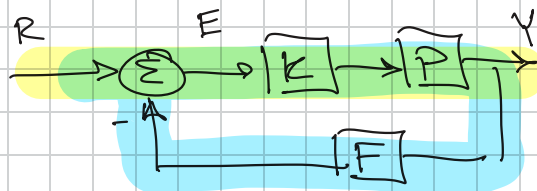
There's a trick:

closed loop
TF
b/n output
and ref

$$\left. \begin{array}{l} \text{closed loop} \\ \text{TF} \\ \text{b/n output} \\ \text{and ref} \end{array} \right\} = \frac{\text{forward}}{1 + \text{loop}}$$



$$\frac{Y}{R} = \frac{G_1}{1 + G_1 G_2}$$



$$\frac{Y}{R} = \frac{K P}{1 + K P F}$$

$$\frac{Y}{E} = \text{exercise}$$

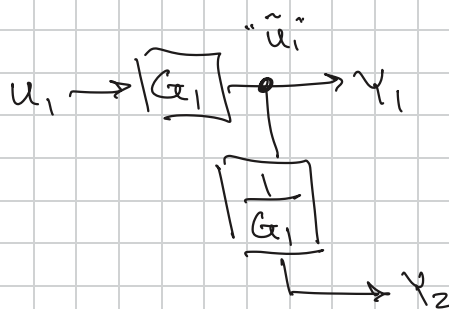
$$\frac{E}{R} = \text{exercise}$$

Moving Blocks Around

①

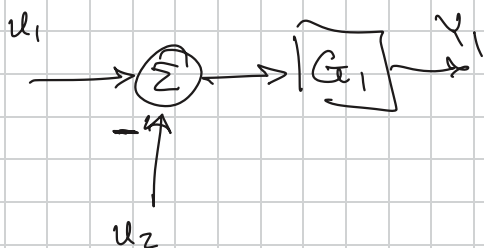


equivalent

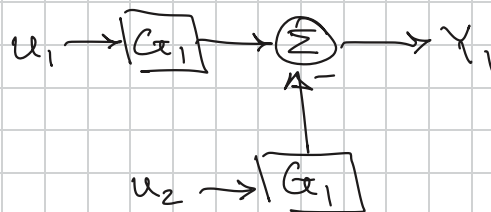


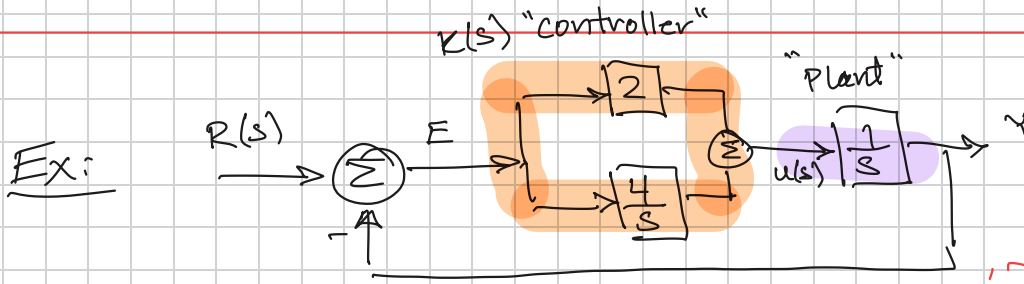
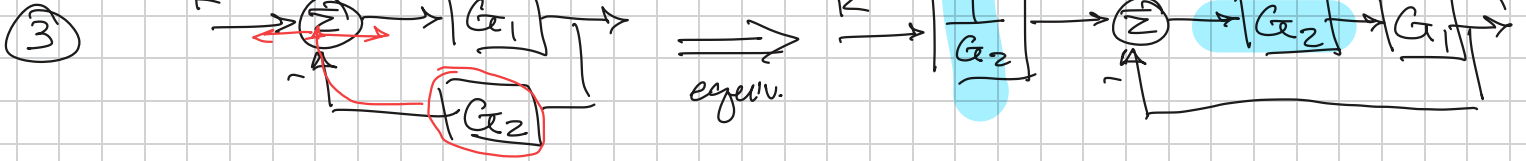
$$\left. \begin{array}{l} y_1 = G_1 u_1 \\ y_2 = u_1 \end{array} \right\} \begin{array}{l} \tilde{u}_1 = u_1 G_1 \\ \Rightarrow y_1 = \tilde{u}_1 \\ \Rightarrow y_2 = \tilde{u}_1 / G_1 \end{array}$$

②



equiv.





Plant: $\frac{Y}{u} = \frac{1}{s}$

signals

$\mathcal{L}\{1(t)\} = 1/s$

$\mathcal{L}\{\delta(t)\} = 1$

$1/s$ is an integrator

$\Rightarrow Y(s) \cdot s = u(s)$

$\Rightarrow \mathcal{L}^{-1}\{Y(s) \cdot s\} = \mathcal{L}^{-1}\{u(s)\}$

$\Rightarrow \dot{y} = u \Rightarrow \int \dot{y} dt = \int u dt$

e.g., $\dot{u} = a = F/m$

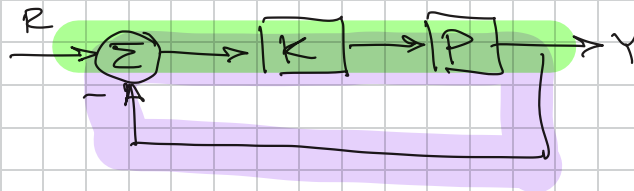
$\dot{p} = v$

Controller: is PI

$\rightarrow K(s) = 2 + \frac{4}{s} = \frac{2 \cdot s + 4}{s}$

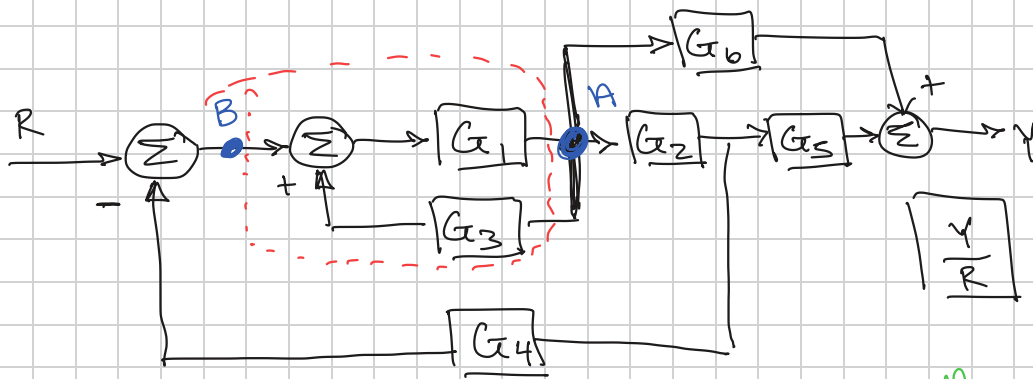
$\frac{u(s)}{E(s)} = K(s)$

and $P(s) = \frac{1}{s}$

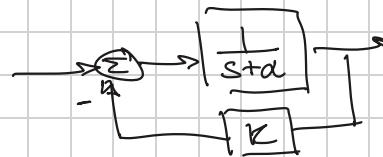
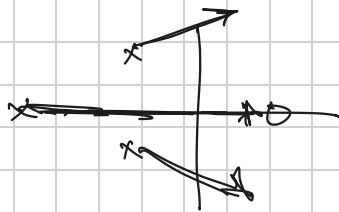
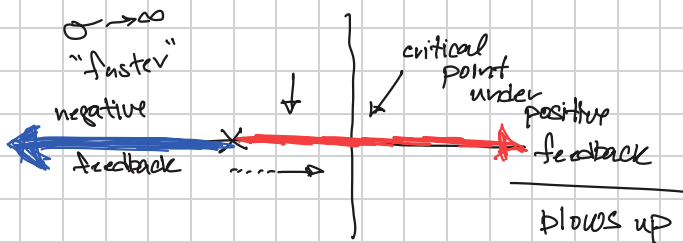
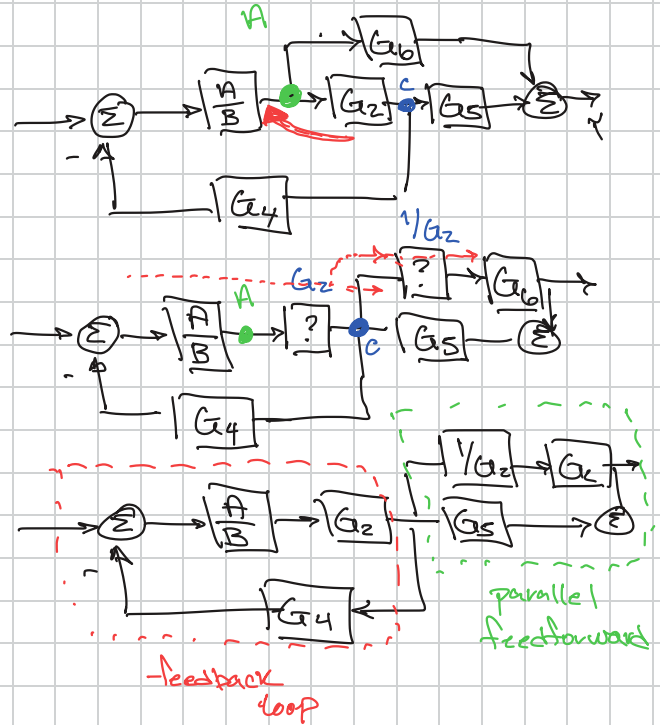


$\frac{Y}{R} = \frac{KP}{1+KP}$

$\frac{Y}{R}$ is the "closed loop" transfer function



$$\frac{A}{B} = \frac{G_1}{1 - G_1 G_3} \Rightarrow$$



Negative Feedback is usually stabilizing
Positive Feedback is usually de-stabilizing

